

Assignment 1

BME.70034: Empirical Asset Pricing

Spring 2026

Instructor Information

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General Rule

- Submit two PDF files:
 - Architecture description (+ repository URL)
 - Results
- Your submission should be self-contained and fully executable — including any data downloads — producing the reported results.
 - Include 'requirements.txt' for the dependency information.
- Your code should include comments that explain the purpose and function of each chunk.
- You are encouraged to discuss this assignment with your classmates.
- You are more than encouraged to use LLM-assisted coding (i.e., vibe coding).
- If needed, obtain asset pricing factors from the Open Source Asset Pricing (<https://www.openassetpricing.com/>) or Ken French Data Library (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html).
- Make logical assumptions where necessary. For additional assumptions beyond what is provided, cite supporting academic papers.
- Copying and pasting another student's assignment is strictly forbidden.
- Read the syllabus and introduction slides carefully for more rules.

Sample

- U.S. common stock monthly returns from CRSP+Compustat between 2017 and 2025.
- Use stocks in NYSE, NYSE American (AMEX), and NASDAQ.
- Usual filter applies (see published papers or codes in Open Source Asset Pricing).
- Winsorize variables (e.g., monthly returns) each month at the top and bottom 1%, if the data contains outliers.

Task

Part A. Data construction

1. Download the monthly time series of the six factors:

$$R_m - R_f, \text{ SMB, HML, RMW, CMA, and MOM}$$

2. Create the CAT factor as:

- A long-short portfolio that is long in stocks with tickers starting with "C" and short in stocks starting with "T".
- Use equal-weighted portfolios with monthly rebalancing.
- Assign stocks based on their ticker as of each month.

3. For each stock, run a time-series regression on the seven-factor model ($R_m - R_f$, SMB, HML, RMW, CMA, MOM, and CAT) to estimate factor loadings.

- Use a rolling window regression. The estimation window is months $t - 24$ to $t - 1$ (strictly past, excluding the current month).
- Drop if there are fewer than 10 months of data in the estimation window.

4. This will give you a stock-month level panel dataset of estimated factor loadings, from 2019 to 2025.

Part B. Regressions

1. Using the panel data, run the following cross-sectional regression:

- Dependent variable: the month- t excess stock return.
- Independent variables: the seven factor loadings estimated from the rolling regression using data strictly before month t .

2. Use the following four specifications:

- (a) Pooled OLS regression.

- (b) Fama–MacBeth regression with Newey–West corrected standard errors (use pre-built functions/packages available).
- (c) Pooled OLS regression with date and stock fixed effects (use pre-built functions/packages available).
- (d) Same as (c), but with standard errors two-way clustered by date and stock.

Part C. GRS Test

1. Sort stocks into 25 portfolios (5×5) based on the estimated CAT loading and one other factor loading of your choice (e.g., HML beta). Use quintile breakpoints each month from the cross-sectional distribution of estimated loadings.
2. Compute equal-weighted monthly excess returns for each of the 25 portfolios.
3. Run a time-series regression of each portfolio's excess return on the seven factors ($R_m - R_f$, SMB, HML, RMW, CMA, MOM, and CAT).
4. Perform the GRS test (Gibbons, Ross, and Shanken, 1989) on the joint null hypothesis that all 25 intercepts (alphas) are zero.
5. Report:
 - (a) The GRS F -statistic and its p -value.
 - (b) The average absolute intercept across the 25 portfolios.
 - (c) A 5×5 heatmap of the intercepts.
6. Discuss: Does the seven-factor model adequately price the cross-section of these 25 portfolios? What does the pattern of intercepts suggest about which dimension of returns the model fails to capture?
7. Repeat Steps 2–6 of Part C, but using the six factors without CAT ($R_m - R_f$, SMB, HML, RMW, CMA, and MOM). Note that the sorting in Step 1 remains the same. Compare the GRS test results across the seven-factor and six-factor models: does including the CAT factor increase/reduce the GRS F -statistic and the average absolute intercept? If so, why?

Expected outputs

1. Regression table for the panel data regressions:
 - Report coefficients, standard errors (in parentheses), number of observations (N), and R^2 .
 - Mark *, **, *** for 10%, 5%, 1% levels of statistical significance, respectively.
 - Present all four specifications (pooled OLS, Fama–MacBeth with Newey–West, pooled OLS with two-way FE, and pooled OLS with two-way FE and two-way clustered s.e.) side by side in a single table with four columns.
2. Summary statistics for the CAT factor:
 - Report the mean, standard deviation, and correlations with the other six factors.
3. Briefly discuss the regression results in the context of validating the seven factors.
4. Briefly discuss similarities or differences across the four specifications and why they arise.
5. GRS test results (separately for the two sets — the six and seven factor models):
 - Report the GRS F -statistic, its p -value, and the average absolute intercept across the 25 portfolios.
 - Include a 5×5 heatmap of the intercepts.
 - Briefly discuss whether the seven-factor model adequately prices the 25 portfolios and what the pattern of intercepts suggests.